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shows that the attempted traceless piece $\left(\mathcal{P}\cdot\partial\epsilon\right)^{A}{}_{BC}$ necessarily vanishes: it is a linear, single-vector construct and is thus killed by $P^{(q)}$ (Cor. B.2). The ansatz collapses to

$$T^{A}{}_{BC} = A\,\delta^{A}{}_{[B}\partial_{C]}\epsilon + B\,\epsilon^{A}{}_{BCD}\partial^{D}\epsilon, \tag{65}$$

recovering Eq. (4.2) of the main text. The scalar projector $\Pi_{\rm PT}[\cdot]$ removes PT-odd *scalars* built from the axial seed (Sec. 2), and the Palatini connection equation then sets $B{=}0$ while fixing $A = 2\eta$ under the trace lock $T_\mu = 3\eta\,\partial_\mu\epsilon$ (Sec. 4.1). This yields the uniqueness map $T^{A}{}_{BC} = 2\eta\,\delta^{A}{}_{[B}\partial_{C]}\epsilon$ quoted in Theorem 4.

Appendix B.6. Consistency Check with the Quadratic Invariant

With the normalization (58) and the C1 map (so $S^\mu{=}0$ and $q_{\lambda\mu\nu}{=}0$),

$$\Pi_{\rm PT}\big[T^{A}{}_{BC}T_{A}{}^{BC}\big] = \frac{2}{3}\,T_\mu T^\mu = \frac{2}{3}\,(3\eta)^2\,\Sigma_\epsilon = 6\,\eta^2\,\Sigma_\epsilon, \qquad \Rightarrow \qquad I_T \equiv -\frac{1}{4}\,\Pi_{\rm PT}\big[T^{A}{}_{BC}T_{A}{}^{BC}\big] = -6\,\eta^2\,\Sigma_\epsilon, \tag{66}$$

as used throughout Secs. 4–5. Here $\Sigma_\epsilon \equiv \Pi_{\rm PT}\big[(\partial\epsilon)^2\big]$ is the *projected, real* scalar, and the sign bookkeeping is carried by $\mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon)$; the unit 1-form $\hat{n}_\mu \equiv \partial_\mu\epsilon/\sqrt{|\Sigma_\epsilon|}$, the canonical traceless rank-one matrix $\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)(\hat{n}^\mu\hat{n}_\nu - \frac{1}{4}\delta^\mu{}_\nu)$, and the trace scale $\tau \equiv \hat{n}^\mu T_\mu = 3\eta\ \mathrm{sgn}(\Sigma_\epsilon)\,\sqrt{|\Sigma_\epsilon|}$ are recalled from Sec. 3.

Appendix B.7. Edge Cases and Patches

On loci where $\Sigma_\epsilon = 0$ the normalized direction $\hat{n}$ is defined patchwise (or by continuity); all algebraic projector statements remain valid, and the conclusions above hold on any patch with $\Sigma_\epsilon \neq 0$. Global/topological subtleties (multi-valued ϵ, nontrivial bundles) lie outside the posture A1–A5 (Sec. 2).

Summary of App. B. We have given explicit, idempotent projectors onto the three torsion irreps, fixed the quadratic identity in the normalization used in the main text, and proved the *single-vector no-go*: from one covector v_μ no nonzero traceless torsion irrep can be built. This reduces the most general one-derivative ansatz to the $\{\delta^{A}{}_{[B}\partial_{C]}\epsilon,\ \epsilon^{A}{}_{BCD}\partial^{D}\epsilon\}$ span, after which the Palatini equations and the scalar projector select the pure-trace map used in the C1 uniqueness theorem.

Appendix C. Appendix C: Three–Chain Reductions & Improvement Currents (σ_ϵ scheme)

Scope (σ_ϵ scheme and naming). This appendix provides the paper–checkable reductions behind Sec. 5 under the *sign-compensated* convention $\mathrm{sgn}(\Sigma_\epsilon)\,I_T \equiv \mathrm{sgn}(\Sigma_\epsilon)I_T = -6\eta^2|\Sigma_\epsilon|$ and the rank-one determinant route naming (formerly "DBI"-type; not Born–Infeld gravity): (i) a rank-one determinant route built out of the canonical traceless matrix $\mathcal{T}^\mu{}_\nu$, (ii) a closed–metric rank–one deformation, and (iii) the PT–even CS/Nieh–Yan shadow. At quadratic order, *each route* reduces in the bulk to the same invariant line

$$\boxed{\delta^2\mathcal{L}_X = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T\ +\ \nabla_\mu J^\mu_X, \qquad A_\star = \lambda^2/8,\ \ X \in \{rank-onedeterminantroute, \mathrm{CM}, \mathrm{CS}^+\}\ ,}$$

with improvements $\nabla_\mu J^\mu_X$ differing by boundary choices (A4/A5). Closed representatives for J^μ are given on FRW/weak–field backgrounds.

Notation and key relation. We use the preamble shorthands $\Sigma_\epsilon \equiv \Pi_{\rm PT}\big[(\partial\epsilon)^2\big]$, $\mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon)$, $\hat{n}_\mu \equiv \dfrac{\partial_\mu\epsilon}{\sqrt{|\Sigma_\epsilon|}}$, $\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat{n}^\mu\hat{n}_\nu - \frac{1}{4}\delta^\mu{}_\nu\Big)$, $\tau \equiv \hat{n}^\mu T_\mu = 3\,\eta\ \mathrm{sgn}(\Sigma_\epsilon)\,\sqrt{|\Sigma_\epsilon|}$ so that $\mathrm{Tr}\mathcal{T} = 0$, $\mathrm{Tr}(\mathcal{T}^2) = \frac{3}{4}$, and (after C1)

$$\tau^2 = 9\eta^2|\Sigma_\epsilon| = \frac{3}{2}\,(-\,\mathrm{sgn}(\Sigma_\epsilon)I_T), \qquad \mathrm{sgn}(\Sigma_\epsilon)\,I_T \equiv \mathrm{sgn}(\Sigma_\epsilon)I_T = -6\eta^2|\Sigma_\epsilon|.$$

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